

Surgery for Anomalous Aortic Origin of a Coronary Artery (AAOCA) in Children and Adolescents: A Meta-Analysis

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Abstract

We conducted a systematic review and meta-analysis of the literature to assess the outcomes of surgery for the anomalous aortic origin of a coronary artery in children and young adults (<30 years). Thirteen publications were selected, including a total of 384 patients. Unroofing of the intramural segment was adopted in 92% (95% CI: 81%-98%) of cases, with pooled early and late mortality of 0% (95% CI: 0%-0.3%) and 0.1% (95% CI: 0%-1%), respectively. Reoperation for aortic regurgitation was anecdotal. Surgical management of anomalous aortic origin of coronary arteries can be achieved with excellent results in pediatric patients, but concerns remain about the durability of surgery.

Keywords

coronary artery anomaly, coronary artery surgery, statistics, meta-analysis, pediatric, congenital heart disease

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Introduction

Anomalous aortic origin of a coronary artery (AAOCA) is a congenital defect that affects <1% of the general population.¹ In particular, the anomalous origin from the opposite aortic sinus has been described to be the second leading cause of sudden cardiac death (SCD) in young athletes.² Timing and indications for surgical correction have been defined by American Association for Thoracic Surgery guidelines in 2016. The European Society of Cardiology (ESC) in 2020^{3,4} reported excellent outcomes at a medium-term follow-up with the majority of patients being free from complications, symptoms, and physical activity restriction.⁵ However, most studies include adult patients only or mixed pediatric-adult series, obscuring tangible outcomes in children and adolescents.⁶

There are several concerns about the optimal management of AAOCA in the pediatric population, and they range from diagnosis and patient selection to surgical indication. In particular, the nonspecific symptoms and, marginally, the suboptimal reliability of noninvasive diagnostic tools and provocative tests available for younger patients,⁷ often delay the time of a correct diagnosis and fail to stratify the clinical risk in such patients accurately. Furthermore, the claimed low risk of SCD under 10 years of age⁸ precludes most patients from consideration for surgical treatment, despite reported death cases in this age group.^{9,10} Last, the more surgical complexity due to smaller anatomic dimensions and the lack of data on long-term follow-up and durability of surgical repair in pediatric patients generate considerable perplexities on timing and which surgical strategy to adopt.^{11,12}

With the rationale of providing a current overview of the clinical outcomes of surgery for AAOCA in children,

adolescents, and young adults (<30 years), we conducted a systematic review and a meta-analysis of scientific literature, with a goal of clarifying the following topics:

- The clinical and anatomical presentation of AAOCA in children and young adults.
- The types of preferred surgical approaches and their outcomes, in terms of early and late mortality and significant complications.
- The risk of development of aortic regurgitation and its association with commissure resuspension or manipulation at the time of AAOCA repair.

In addition, we performed a sub-analysis focused on pediatric patients (<18 years) only.

Materials and Methods

Data Collection

A systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses

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Abbreviations

AAOCA	anomalous aortic origin of a coronary artery
AAOLCA	anomalous aortic origin of left coronary artery
AAORCA	anomalous aortic origin of right coronary artery
CABG	coronary artery bypass grafting
SCD	sudden cardiac death
SD	standard deviation

statement protocol.¹³ This study was prospectively registered on the PROSPERO database (registration number CRD42021275235, August 2021). The PubMed, Web of Science, and Scopus databases were systematically searched in October 2021 to identify all published studies investigating the clinical outcomes of surgical treatment of AAOCA in the pediatric population. Search string: ([anomalous aortic origin coronary artery] OR [aaoca]) AND ([pediatric] OR [children]). During the study selection by two authors (MP and MAP), any initial disagreement regarding eligibility was resolved by discussion among all the authors and then agreement by consensus.

Inclusion Criteria. After elimination of duplicates, all studies underwent the first screening on the title and abstract and subsequent full-text review, using the following inclusion criteria: (1) prospective and/or retrospective observational reports describing surgical repair of AAOCA; (2) pediatric patients or young adults (mean/median age <18 years and maximal age <30 years); (3) studies written in English after 1968; (4) series with 5 or more enrolled patients.

Exclusion Criteria. Exclusion criteria were: (1) duplicates; (2) results from not full-text articles; (3) studies not written in English; (4) mean/median age of patients at study >18 years or maximal age >30 years; (5) case reports and series with <5 patients; (6) reviews and meta-analyses; and (7) series without surgical treatment or outcomes.

Data Extraction. Two authors extracted data on study characteristics and results to a preset Excel abstraction form. Extracted data were: publication year, the number of patients, mean/median age at operation, mean/median follow-up period, gender, cohort period, anatomical distribution of AAOCA, pre and postoperative symptoms of possible cardiac origin (chest pain, dyspnea, syncope, and palpitations at rest or during exercise), aborted preoperative and postoperative SCD, type of surgical treatment, early and late mortality, reoperations, development of moderate or more aortic regurgitation, postoperative pericardial effusions, and exercise restriction (defined as restriction from all competitive sports and physical exercise with a moderate or high dynamic component). Outcomes were presented as pooled proportions for data synthesis.

Quality Assessment. The methodological quality of the study was evaluated by the critical appraisal tool for cross-sectional studies (AXIS)¹⁴ by 2 authors. Disagreements between the 2

authors on reports' selection were resolved by consensus with the help of a third reviewer.

Statistical Analysis. Data are presented descriptively as mean $\pm$ standard deviation (SD) or median (range) in the case of quantitative variables, and as counts and percentages in the case of categorical variables. Fixed and random-effects meta-analysis for pooled proportion was performed with R version 4.0.4 patched using the "metaprop," "forest," and "funnel" functions in "meta" package by age populations (pediatric and pediatric + young adults) and for the combined ones. The proportions were arcsine transformed; the between-study variance tau-squared was estimated with the DerSimonian-Laird method; the confidence intervals for individual study results were calculated with the Clopper-Pearson method; the pooling of the studies was done with the inverse method. The forest plots show the results of individual and pooled analyses of included studies. The heterogeneity of the study results among the included studies was assessed using I²-index statistics and *p*-value for Cochran Q statistics. The Cochran Q statistic was considered significant at *p* < .05; an I² value of >50% indicated substantial heterogeneity. A random-effect model is adopted in the case of substantial heterogeneity, and clinical and methodological heterogeneity is addressed. Subgroup analyses are undertaken, considering the age of patients and distinguishing studies with exclusively pediatric patients (maximal age <18 years) and mixed pediatric and young adult patients.

Results

After the elimination of duplicates, a total of 914 reports were identified through database search, and full-text eligibility was assessed for 50 of them. Twenty-nine manuscripts described outcomes in nonpediatric cohorts and 7 did not provide results of surgical treatment of AAOCA and were excluded. Due to a possible overlap of patients, the 2020 Congenital Heart Surgeons' Society AAOCA registry study by Jegatheeswaran et al¹⁵ was excluded from this meta-analysis. However, given its significance in the literature, we summarized its findings in Tables 1 to 3. Finally, 13 articles were included in this review^{16–28} (Figure 1 and Table 1). Figure 2 summarizes the quality assessment of selected reports using AXIS.

Patient Characteristics

We identified 384 patients, with a mean/median age ranging from 10.5 to 16.5 years. Anomalous aortic origin of left coronary artery (AAOLCA) was present in 105 of them (27%), while the anomalous aortic origin of right coronary artery (AAORCA) was in 279 (73%). The prevalence range of AAOLCA across series was 16%–100%, while it was 0%–84% for AAORCA. When reported, an interarterial course was observed in all but 2 series. The prevalence of intramural origin ranged between 33% and 100% of patients. Patients

Table 1. Included Studies and Demographic Characteristics of Patients.

Selected for meta-analysis	First author	Year	Cohort period	Age at surgery, years		Male	Follow-up, years		AAOLCA		AAORCA		AAOLCA + AAORCA		Interarterial		Intramural		Intraconal		Symptomatic		Aborted SCD			
				median (range)/ mean $\pm$ SD	n (%)		Median (range)/ mean $\pm$ SD	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)
Pediatric only	Osaki	2007	7	1994-2006	10.5 (2.7-14.5)	6 (86%)	1.9 (1.2-3)	0	7 (100%)	0	0	0	7 (100%)	—	—	7 (100%)	—	—	7 (100%)	—	7 (100%)	—	1 (14%)			
	Wittlieb-Weber	2014	24	2001-2007	12 (5-18)	17 (71%)	5.3 (1-9.2)	8 (33%)	16 (67%)	0	0	0	24 (100%)	24 (100%)	0	24 (100%)	24 (100%)	0	17 (71%)	0	17 (71%)	1 (4%)				
	Herrmann	2016	20	2000-2014	16.5 (14-18)	17 (85%)	In-hospital mortality	7 (35%)	13 (65%)	0	0	0	—	—	—	—	—	—	14 (70%)	—	—	—				
	Balasubramanya	2016	66	2006-2014	12.4 $\pm$ 4	41 (62%)	5.7 $\pm$ 2.2 (median 5.6)	11 (17%)	55 (83%)	0	0	0	66 (100%)	66 (100%)	0	66 (100%)	66 (100%)	0	52 (79%)	0	52 (79%)	—				
	Sachdeva	2017	63	1999-2015	13 (0.5-18)	43 (68%)	3.1 (0-13.6)	13 (21%)	50 (79%)	0	0	0	63 (100%)	63 (100%)	0	63 (100%)	63 (100%)	0	34 (54%)	0	34 (54%)	4 (6%)				
	Mery	2018	44	2012-2017	14 (8-18)	31 (70%)	2 (0-4)	9 (20%)	35 (80%)	0	0	0	—	—	—	41 (93%)	—	—	32 (73%)	—	32 (73%)	3 (7%)				
	Bonilla-Ramirez	2021	61	2012-2019	13 (11-15)	39 (64%)	4 (2-6)	10 (16%)	51 (84%)	0	0	0	60 (98%)	60 (98%)	0	60 (98%)	60 (98%)	0	—	—	—	2 (3%)				
	Frommelt	2003	8	1997-2002	13 $\pm$ 4.7	6 (75%)	1.5 (median)	5 (63%)	3 (37%)	0	0	0	8 (100%)	8 (100%)	0	8 (100%)	8 (100%)	0	4 (50%)	0	4 (50%)	1 (13%)				
	Erez	2006	9	2003-2005	12 $\pm$ 5.8 (0.3-23)	8 (89%)	1 $\pm$ 0.7 (0.3-2.5)	4 (44%)	5 (56%)	0	0	0	9 (100%)	3 (33%)	0	9 (100%)	3 (33%)	0	8 (89%)	0	8 (89%)	0				
	elZein	2009	8	2002-2007	11.7 $\pm$ 11.1 (0.2-28)	6 (75%)	1.2 $\pm$ 1.3 (0.1-3.3)	2 (25%)	6 (75%)	0	0	0	8 (100%)	8 (100%)	0	8 (100%)	8 (100%)	0	4 (50%)	0	4 (50%)	0				
Pediatric and young adults	Brothers	2015	9	2009-2015	14.7 (5-19)	9 (100%)	0.7 (median 0.5)	2 (23%)	7 (77%)	0	0	0	9 (100%)	9 (100%)	0	9 (100%)	9 (100%)	0	8 (89%)	0	8 (89%)	0				
	Yerebakan	2019	26	2004-2016	15.5 (IQR 10.2-17.7)	—	4.3 (IQR 3.5-5)	16 (62%)	10 (38%)	0	0	0	26 (100%)	26 (100%)	0	26 (100%)	26 (100%)	0	20 (77%)	0	20 (77%)	2 (8%)				
	Mostefa Kara	2020	39	2005-2019	14 (10-26)	30 (77%)	4 (3-6)	11 (28%)	28 (72%)	0	0	0	26 (67%)	28 (72%)	-	26 (67%)	28 (72%)	-	31 (80%)	0	31 (80%)	2 (5%)				
	Jegatheeswaran	2019	395	2000-2018	13.3 (0.9-30.7)	261 (66%)	2.8 (0-16.2)	108 (27%)	282 (72%)	5 (1%)	5	5	382 (97%)	369 (93%)	7 (2%)	382 (97%)	369 (93%)	7 (2%)	-	—	—	3 (1%)				

Abbreviations: SD, standard deviation; AAOLCA, anomalous aortic origin of left coronary artery; AAORCA, anomalous aortic origin of right coronary artery; SCD, sudden cardiac death; IQR, interquartile range.

Table 2. Types of Surgical Treatment Across Series.

Selected for meta-analysis	Pediatric only	First author	Year	Unroofing n (%)	Osteoplasty n (%)	Reimplantation n (%)	CABG n (%)	Pulmonary translocation n (%)	Other procedures n (%)	Commissure manipulation n (%)
		Osaki Wittlieb- Weber	2007	5 (71%)	0	2 (29%)	0	0	0	—
			2014	23 (96%)	0	1 (4%)	0	0	0	16 (70%)
		Herrmann	2016	19 (95%)	0	1 (5%)	0	0	0	0
		Balasubramanya	2016	66 (100%)	0	0	0	0	0	1 (1.5%)
		Sachdeva	2017	63 (100%)	0	0	0	0	0	5 (8%)
		Mery	2018	35 (80%)	1 (2%)	7 (16%)	0	0	1 (2%)	—
		Bonilla-Ramirez	2021	45 (74%)	0	16 (26%)	0	0	0	0
		Frommelt	2003	7 (88%)	0	0	0	0	1 (12%)	4 (50%)
		Erez	2006	2 (22%)*	0	6 (67%)*	0	0	0	6 (67%)
		elZein	2009	8 (100%)	0	0	0	0	0	2 (25%)
Pediatric and young adults		Brothers	2015	9 (100%)	0	0	0	0	0	2 (22%)
		Yerebakan	2019	26 (100%)	0	0	0	0	0	17 (65%)
		Mostefa Kara	2020	30 (77%)	0	2 (5%)	3 (8%)	3 (8%) [isolated]; 11 (28%) [+unroofing]	1 (3%)	3 (8%)
		Jegatheeswaran	2019	344 (87%)	25 (6%)	24 (6%)	3 (1%)	22 (6%)	14 (4%)	104 (26%)

Abbreviation: CABG, coronary artery bypass grafting.

*1 patient awaits surgery.

Table 3. Late Outcomes of Surgery at Last Available Follow-up.

Selected for meta-analysis	Pediatric only	First author	Year	Early mortality		Late mortality	Follow-up, years Median (range)/ mean $\pm$ SD	Reoperation n (%)	Reoperation for aortic Regurgitation, n (%)	Heart transplantation n (%)	$\geq$ Moderate aortic Regurgitation, n (%)	Symptomatic n (%)	Aborted SCD n (%)	Physical activity Restriction, n (%)
				n (%)	n (%)									
Pediatric only		Osaki	2007	1 (14%)	0	0	1.9 (1.2-3)	0	0	0	—	1 (14%)	0	—
		Wirtlieb-Weber	2007	0	0	0	5.3 (1-9.2)	2 (8%)	0	0	0	13 (54%)	0	0
		Herrmann	2016	0	—	—	In-hospital mortality	—	—	—	—	—	—	—
		Balasubramanya	2016	0	0	0	5.7 $\pm$ 2.2	1 (1.5%)	0	0	—	—	—	—
		Sachdeva	2017	0	1 (1.6%)	1 (1.6%)	3.1 (0-13.6)	2 (3%)	1 (1.6%)	1 (1.6%)	1 (1.6%)	23 (37%)	2 (3%)	0
		Mery	2018	0	0	0	2 (0.1-4)	3 (7%)	0	0	—	4 (9%)	1 (2.3%)	2 (4.5%)
		Bonilla-Ramirez	2021	0	0	0	4 (2-6)	2 (4%)	0	0	0	7 (12%)	1 (2%)	4 (7%)
		Frommelt	2003	0	0	0	1.5 (median)	0	0	0	0	0	0	0
		Erez	2006	0	0	0	1 $\pm$ 0.7 (0.3-2.5)	0	0	0	—	0	0	0
		elZein	2009	0	0	0	1.2 $\pm$ 1.3 (0.1-3.3)	0	0	0	—	0	0	—
		Brothers	2015	0	0	0	0.7 (median 0.5)	—	—	—	0	—	—	—
		Yerebakan	2019	0	1 (3%)	1 (3%)	4.3 (IQR 3.5-5)	1 (3%)	1 (3%)	0	3 (12%)	0	0	0
		Mostefa Kara	2020	0	0	0	4 (3-6)	2 (5%)	0	0	0	—	—	0
		Jegatheeswaran	2019	4 (1%)	0	0	2.8 (0-16.2)	30 (8%)	2 (0.5%)	1 (0.3%)	7 (2%)	—	—	—

Abbreviations: SD, standard deviation; SCD, sudden cardiac death; IQR, interquartile range.

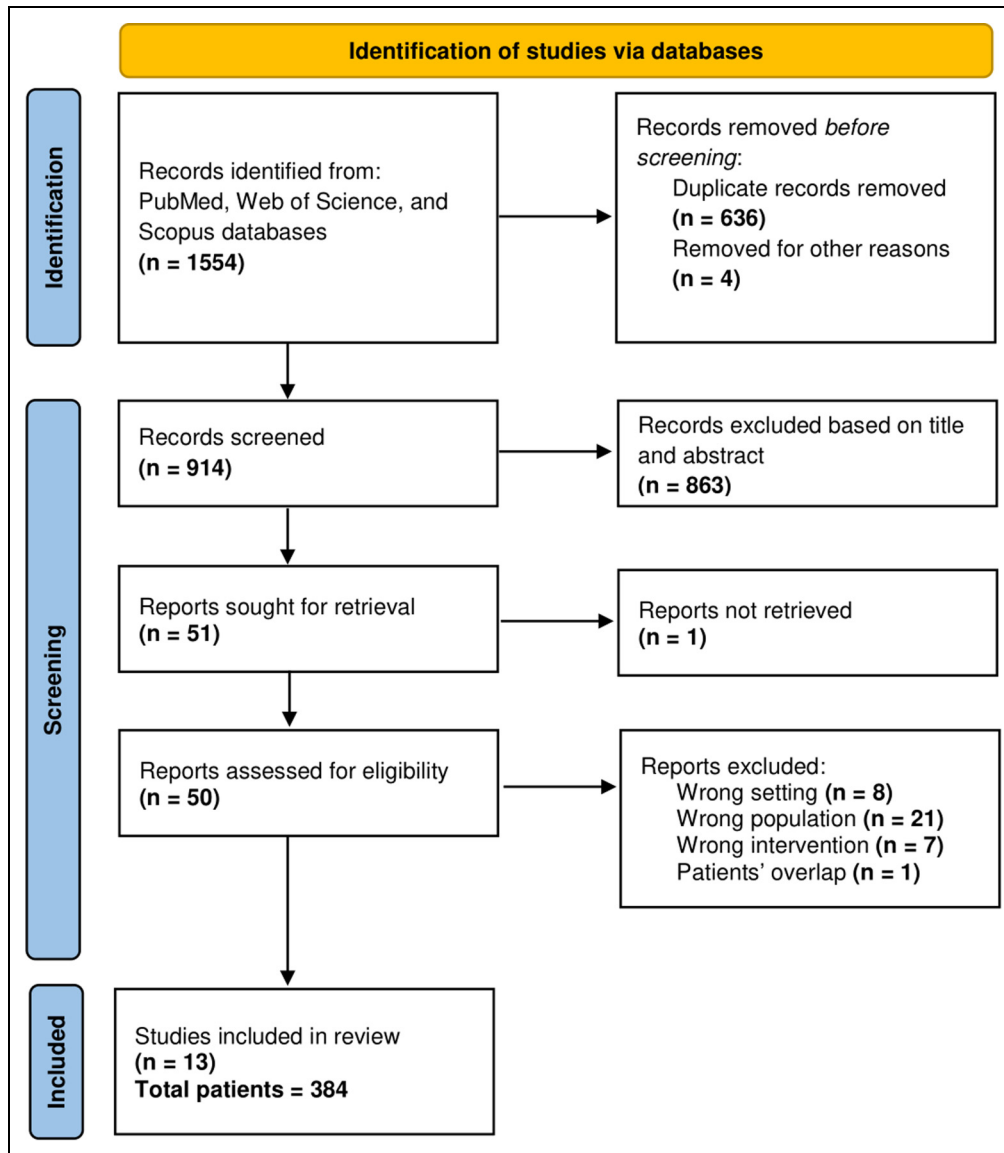


Figure 1. Preferred reporting items for systematic reviews and meta-analyses (PRISMA) 2020 flow diagram for new systematic reviews.

were symptomatic in 50% to 100% of cases, with a percentage of aborted SCD at presentation up to 14% (see Table 1).

manipulation was reported in 21% (95% CI: 6%-40%) of the cases (Figure S1, $I^2 = 93\%$, 95% CI: 90%-96%, $p < .0001$).

Surgical Treatment

From a meta-analysis of the selected studies, unroofing of the intramural segment was the most frequently performed procedure, with 92% (95% CI: 81%-98%) of patients treated with this technique (Figure S1, $I^2 = 89\%$, 95% CI: 83%-93%, $p < .0001$), followed by coronary ostium reimplantation with a pooled prevalence of 5% (95% CI: 1%-14%, Figure S1, $I^2 = 87\%$, 95% CI: 79%-91%, $p < .001$). Osteoplasty, coronary artery bypass grafting (CABG), pulmonary artery translocation, and other procedures were adopted less frequently (see Table 2). Of note, aortic commissure resuspension/

Outcomes

There were 1 early (<30 days or in-hospital) and 2 late deaths across all series, at a median follow-up time ranging from 0.5 to 5.6 years (Table 3). The causes of death were postoperative cardiogenic shock in one,¹⁶ SCD with ventricular fibrillation in another,²⁰ and endocarditis after mechanical aortic valve replacement for development of significant postoperative aortic regurgitation in the last case.²⁷ Our meta-analysis showed a pooled early mortality of 0% (95% CI: 0%-0.3%) (Figure S2, $I^2 = 0.0\%$, 95% CI not estimable, $p = .981$), and a pooled late mortality of 0.1% (95% CI: 0%-1%) (Figure S2, $I^2 = 0\%$, 95% CI 0%-26%, $p = .860$). Cardiac reoperations for

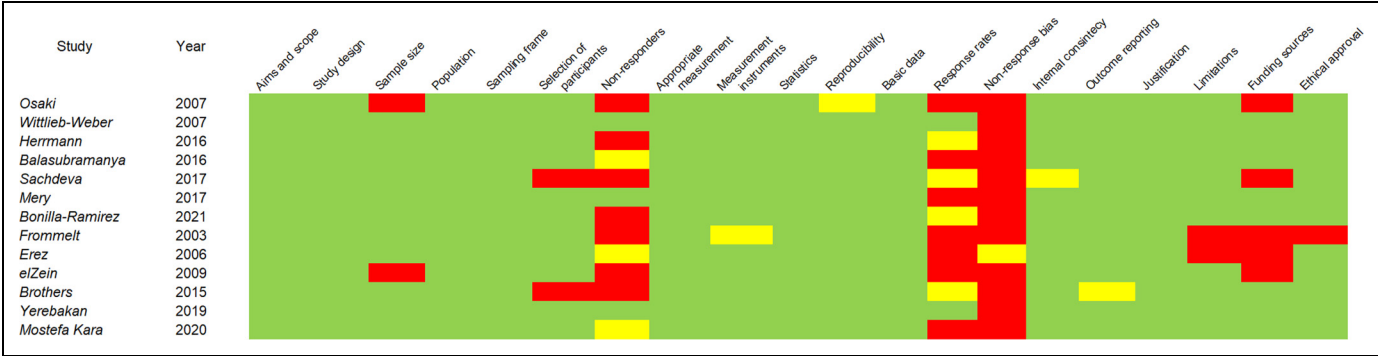


Figure 2. Summary of quality assessment of reports using AXIS. Green indicates AXIS criteria fully satisfied; yellow indicates AXIS criteria partially satisfied; red indicates AXIS criteria not satisfied.

all causes during follow-up occurred in 3% (95% CI: 2%-5%) of patients (Figure S2, $I^2 = 0\%$, 95% CI: 0.0%-49%, $p = .643$). In particular, need for heart transplantation was anecdotal, affecting 0.1% (95% CI: 0.0%-1%) ($I^2 = 0.0\%$, 95% CI not estimable, $p = .973$), as well as reoperation for aortic valve regurgitation in 0.1% (95% CI: 0%-1%) (Figure S2, $I^2 = 0.0\%$, 95% CI: 0%-35%, $p = .803$). At last follow-up, 9% (95% CI: 1%-23%) of patients are symptomatic ($I^2 = 88\%$, 95% CI: 80%-93%, $p < .001$), and recurrence of aborted SCD occurred in 1% (95% CI: 0%-3%) of cases ($I^2 = 0\%$, 95% CI: 0%-49%, $p = .702$). Only 1% (95% CI: 0%-3%) of patients are restricted from physical activity ($I^2 = 54\%$, 95% CI: 0%-79%, $p = .03$).

Patients Under 18 Years of Age

Seven reports described outcomes in patients <18 years,¹⁶⁻²¹ for a total of 285 children and teenagers (mean/median age: 10.5-16.5 years). When reported, an interarterial course was observed for almost all patients, with intramurality in 93% to 100% of cases. From 54% to 100% of patients were symptomatic (Table 1). Unroofing was the most frequently performed procedure in all series, with a pooled prevalence of 93% (95% CI: 79%-99%), followed by coronary reimplantation in 7% (95% CI: 1%-19%) of patients. Resuspension or manipulation of aortic commissure was performed in 7% (95% CI: 0%-27%) of the patients (Figure S1).

In this cohort, only 1 early death and 1 late death occurred, at a mean/median time of 1.9 to 5.7 years of follow-up, resulting in 0.0% (95% CI: 0.0%-0.4%) and 0.2% (95% CI: 0.0%-1%) early and late mortality (Figure S2), respectively. Reoperation was necessary for 3% (95% CI: 2%-6%) of patients. Only 1 patient developed significant aortic regurgitation, which required surgical treatment. Finally, at the last follow-up, 24% (95% CI: 9%-42%) of patients remain symptomatic after surgery and 2% (95% CI: 0%-7%) are restricted from physical activity.

Discussion

We performed a systematic review and a meta-analysis that summarized the clinical outcomes and the anatomical and surgical features of patients younger than 30 years undergoing

surgery for AAOCA. To the best of our knowledge, this study, which includes 384 patients, is one of the largest surgical cohorts of young patients affected by AAOCA currently available.

It is now evident that the diagnosis of AAOCA may have a negative effect on the quality of life of either the patient, the parents, or both, impacting families' functioning.²⁹ Also, medical management often is associated with the alteration of parents' perception of their child's health.³⁰ For these reasons, consideration of surgical management of AAOCA is currently expanding toward younger ages.

From our literature review, the median age at surgery was lower than 15 years in most cases. AAOCA is often treated in early adolescence in pediatric cardiac surgery centers when the beginning of competitive physical activity unmasks the first occurrence of symptoms or clinical detection of suspicious cardiac signs. About two-thirds of patients were symptomatic, although symptoms were nonspecific in the first instance in most cases. Although rare, the first presentation of AAOCA may be an aborted SCD, and, noticeably, all exclusively pediatric series had a not negligible percentage of this occurrence, which ranged from 3% to 14% (see Table 1). For this reason, the concept of a 10-year-old threshold for clinical and therapeutic consideration of AAOCA may induce an underestimation of the impact of this condition in children.

The clinical presentation of AAOCA can be dramatic at any age, with autopsy series including mostly young patients at the beginning of their competitive physical activity.^{9,10} Thus, the clinical suspicion must be high in children, with subsequent referral to a dedicated pediatric cardiac center. When appropriate, cardiac magnetic resonance imaging, which has been advocated to have the potential to become a helpful screening tool for young athletes,³¹ should be adopted promptly.

As often reported, AAORCA is by far more common than AAOLCA,³² although data are frequently derived from interventional cardiology or surgical registries or autopsy series. Our analysis confirms that in most recent reports, the "surgical" prevalence of AAORCA is 2 to 3 times higher than AAOLCA in pediatric age (Table 1), reflecting both the true prevalence of the disease and the anatomical and clinical indications for surgery adopted by each institution.

Many innovative and refined diagnostic tools (namely virtual angioscopy of magnetic resonance and computer-tomography images, with a 3-dimensional reconstruction of coronary anatomy) have been proposed to help stratify patients' clinical risk with AAOCA.³³⁻³⁵ To date, the finding of interarterial, especially intramural, course is the most reliable anatomical characteristic to guide surgical approach,³⁶ together with the involvement of the left coronary artery and the presence of symptoms or signs of myocardial ischemia, as stated by the most recent ESC guidelines for managing adults with congenital heart disease.⁴ Our data support that an intramural course, which was observed in almost all younger patients (Table 1), is one of the leading factors for surgical referral of patients. Among surgical techniques, unroofing of the intramural segment turned out to be the most frequently employed surgical approach in our literature review (with a pooled prevalence of 92%, 95% CI: 81%-98%). Coronary ostium reimplantation was the second most frequent procedure, while CABG, pulmonary translocation, and other techniques have been described only anecdotally in this population (Table 2). It is of note that utilization of different surgical techniques may depend on surgical skills and backgrounds (congenital vs. adult centers), which may account partially for the high degree of heterogeneity we observed between reports. Thus, a subgroup analysis further distinguishing studies by the age of patients was mandatory to minimize heterogeneity bias.

Consequently, institutions with a predominantly congenital experience adopted unroofing and coronary reimplantation procedures.¹⁶⁻²¹ On the other hand, centers where adult patients are referred may use other techniques, such as osteoplasty or CABG, whose long-term efficacy, especially in young adults, has yet to be determined.²²⁻²⁸ Our findings align with the ongoing Congenital Heart Surgeons' Society registry of AAOCA,¹⁵ which confirms a heterogeneous surgical approach to AAOCA due to the inclusion of pediatric and adult cardiac surgery centers.

Outcomes of surgery for AAOCA are excellent, with 12 of 13 and 10 of 12 series with absent early and late mortality, respectively. When reported, mortality was extremely low (3 deaths among 384 treated patients, 0.78%). As a result, our systematic review suggests that surgical treatment of AAOCA could also be pursued in selected younger patients safely. Particular attention should be paid to successfully correct this anomaly directly because reoperations could be extremely challenging and result in adverse events.¹⁵

Emergent presentation of this condition carries an apparent increased risk of mortality that is not usually related to surgery and may be minimized only with a prompt diagnosis and rapid, effective management, including mechanical cardiac support and then corrective surgery.

The limited follow-up time of every study does not allow speculations about the long-term durability of surgical results. The primary concern that has been advocated by several authors^{12,26} is the occurrence of significant aortic regurgitation following aortic valve commissure

resuspension or surgical manipulation of the aortic root. The literature has not been conclusive yet in determining the real impact of commissure resuspension on aortic valve function. Most reports describe a consistent risk of postoperative regurgitation,^{11,25} recommending avoiding this procedure whenever possible. On the other hand, Yerebakan et al,²⁷ who resuspended the aortic commissure at the mid-point of the unroofed segment routinely (to restore the original height of the commissure, which lowers during the unroofing operation), reported an even lower incidence of novel aortic regurgitation with this approach. From our analysis, commissure manipulation was considered necessary for 21% of patients, especially in studies with older cohorts. In comparison, it is adopted in only 7% of pediatric patients (Table 2), in whom a more conservative surgery seems to be the preferred choice. Interestingly, very few patients develop moderate or more aortic regurgitation at follow-up. It results in a pooled prevalence of reoperation for aortic surgery of 0.1% (95% CI: 0%-1%).

To date, the onset of novel moderate or more aortic regurgitation is rare. It seems to be independent of the frequency of resuspension of aortic commissures. However, data and follow-up times are insufficient to support a more restricted or broader use of this technique and to speculate about the long-term fate of aortic valve competence in those patients who develop novel mild aortic regurgitation following surgery. We suggest that a selective and anatomically guided approach might be optimal for these patients, especially the younger ones where long-term aortic valve function preservation is a primary endpoint of surgery. Encouraging early outcomes have been reported with the transection and reimplantation technique. Herein is the rationale for avoiding aortic valve manipulation and allows for complete elimination of intramural segments, especially when the anomalous coronary course is below the commissure.²² However, the long-term effects of circumferential anastomosis on coronary anatomy are still to be investigated. Results from ongoing prospective and multicenter registries may resolve this issue.^{15,37}

Last, in children and adolescents, the perception of successful surgery is driven mainly by freedom from residual symptoms and restrictions on physical activity. Of note, surgery allowed almost all patients to suspend their physical activity restrictions (Table 3), minimizing personal and social issues. Conversely, exclusively pediatric series report a constant percentage of patients who remain symptomatic after surgery, reaching a pooled prevalence of 24% (95% CI: 9%-42%). We speculate that these numbers could overestimate the actual incidence of true cardiac symptoms after treatment because of the difficulty of discriminating real symptoms for younger patients and the proven absent association between subjective symptoms and objective tests of myocardial perfusion.¹⁷ Dobutamine-stress cardiac magnetic resonance imaging is emerging as a safe and reliable noninvasive diagnostic technique^{38,39} to unmask inducible myocardial ischemia in patients with subtle or even absent clinical manifestations,⁴⁰

representing a promising tool for the management of this challenging population.

Limitations

Our study presents some limitations. Concerns have been reported in performing meta-analyses of nonrandomized clinical trials; however, randomized clinical trials about AAOCA are not available in the scientific literature, representing an intrinsic limitation of current knowledge about AAOCA. Heterogeneity between studies was high, especially when analyzing the preferred surgical strategy. The rarity of a standardized diagnostic and therapeutic approach across institutions might account for this limit. Clinical variation between reports was predominantly related to the age of patients, unmasking different institutional surgical backgrounds, which may represent a bias when analyzing outcomes.

For this reason, we performed a subanalysis on exclusively pediatric patients, and we preferred a random-effect model to minimize heterogeneity bias. The low number of studies satisfying inclusion criteria and the small incidence of postoperative variables did not allow us to restrict selection to more homogeneous reports without compromising the statistical power of analyses. Finally, manuscripts which stratified clinical outcomes according to the need for commissure resuspension were too rare for specific subgroup analysis.

Conclusions

Surgical repair of AAOCA in the pediatric population can be achieved with excellent results. Unroofing of the intramural segment is the most frequently performed procedure in most centers, with reported minimal early and late mortality, in almost all series. The surgery allowed most patients to terminate physical activity restrictions, despite a not negligible percentage of patients who persist symptomatic after surgery. At a mean/median follow-up of 0.5 to 5.6 years, severe aortic regurgitation needing surgical repair seems to be an infrequent occurrence. A longer follow-up is mandatory to assess the actual durability of surgery in this population.

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